

MAT 017B Quiz 2

August 9, 2018

Name: Key

SID: _____

1. Evaluate the following indefinite integrals.

a. $\int 3e^{1-x} dx$

$$u = 1 - x$$

$$du = -dx$$

$$\int 3e^{1-x} dx = \int -3e^u du$$

$$= -3e^u + C$$

$$= \boxed{-3e^{1-x} + C}$$

b. $\int \sqrt{x+3} dx$

$$u = x + 3$$

$$du = dx$$

$$\int \sqrt{x+3} dx = \int \sqrt{u} du$$

$$= \frac{2}{3} u^{3/2} + C$$

$$= \boxed{\frac{2}{3} (x+3)^{3/2} + C}$$

c. $\int \cos x e^{\sin x} dx$

$$u = \sin x$$

$$du = \cos x dx$$

$$\int \cos x e^{\sin x} dx = \int e^u du$$

$$= e^u + C$$

$$= \boxed{e^{\sin x} + C}$$